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Certificate of Calibration

Scan to Authenticate

Calibration of:	Portable Vibration Calibrator
Manufacturer:	The Modal Shop
Model number:	9110D
Serial number:	11755
Calibrated for:	WEARCHECK No 55 Angus Crescent EDEN GLEN EXT 57 Johannesburg Gauteng South Africa
Location of calibration:	NMISA (Pretoria)
Calibration procedure:	AV\VS-0006
Date(s) of calibration:	2026/02/16 to 2026/02/18
Next calibration date:	2027/02/19

Calibrated by: Vumile Tyalimpi

Checked by: Chris Matthee

Technical Signatory:

Signed by:VUMILE MIKE TYALIMPI
Signed at:2026-02-17 14:32:26 +02:00
Reason:I approve this document

Vumile Tyalimpi

For Chief Executive Officer:

Signed by:Oelof Abraham Kruger
Signed at:2026-02-17 15:26:17 +02:00
Reason:I approve this document

Oelof Kruger

1 PROCEDURE

The instrument was calibrated according to the manufacturer's specification using reference accelerometer. The results of the calibration are traceable to the relevant national measurement standards. The following equipment was used:

1.1	Laboratory Standard	Make	Model	Serial No
	Reference Accelerometer	PCB	J353B04	172370

1.2	Other Equipment	Make	Model	Serial No
	Calibration system	SPEKTRA	SRS 35	201719

1.3 Vibration Calibration Software

Manufacturer: SPEKTRA GmbH Dresden
Name: CS18
Version: 2.15.13.0

1.4 Mounting Conditions and Considerations

Mounting Torque : 2 N·m
Lubrication used : Petroleum Jelly
Mounting Orientation : Vertical
Cable : Gentle bend supported by laboratory clamps
Reference Level : At the bottom of the accelerometer
g_n : 9,807 m/s^2

2 RESULTS

2.1 Frequency Response

UUT Applied Values		Measured Output			*U*
Frequency	Level	Frequency	Level		
(Hz)	(m/s²)	(Hz)	(m/s²)	(g_n)	(%)
10	5	10,00	4,85	0,50	3,8
16	5	16,00	4,98	0,51	3,8
20	10	20,01	9,93	1,01	3,8
25	10	25,01	9,94	1,01	3,8
32	10	32,01	9,95	1,01	3,8
50	20	50,02	20,00	2,04	3,8
63	20	63,02	19,97	2,04	3,8
80	20	80,03	19,95	2,03	3,8
100	20	100,0	19,92	2,03	3,8
125	20	125,1	19,97	2,04	3,8
160	20	160,1	19,96	2,04	3,8
200	20	200,1	19,93	2,03	3,8
500	20	500,2	19,94	2,03	3,8

UUT Applied Values		Measured Output			*U*
Frequency (Hz)	Level (m/s²)	Frequency (Hz)	Level (m/s²)	Level (g_n)	(%)
630	20	630,3	20,01	2,04	3,8
800	20	800,3	20,01	2,04	3,8
1 000	20	1000,4	20,07	2,05	3,8
1 250	20	1250,1	20,09	2,05	3,8
1 600	20	1600,4	20,11	2,05	3,8
2 500	20	2501,0	20,10	2,05	3,8
4 000	20	4001,4	20,22	2,06	3,8
5 000	20	5001,9	20,39	2,08	3,8
8 000	20	8003,1	20,86	2,13	3,8
10 000	20	10003,8	21,49	2,19	3,8

2.2 Acceleration Range Accuracy

UUT Applied Values		Measured Output			*U*
Frequency (Hz)	Level (m/s²)	Frequency (Hz)	Level (m/s²)	Level (g_n)	(%)
160	5	160,06	4,97	0,50	3,5
160	10	160,06	9,96	0,51	3,5
160	15	160,06	14,95	1,01	3,5
160	20	160,06	19,93	1,01	3,5
160	50	160,06	49,94	1,01	3,5
160	100	160,06	99,60	2,04	3,5
160	130	160,06	129,50	2,04	3,5
160	150	160,06	149,43	2,03	3,5

2.3 Velocity Range Accuracy

UUT Applied Values		Measured Output		*U*
Frequency (Hz)	Level (mm/s)	Frequency (Hz)	Level (mm/s)	(%)
160	2	160,06	1,99	3,6
160	5	160,08	4,97	3,6
160	10	160,05	9,96	3,6
160	20	160,06	19,94	3,6
160	30	160,06	29,90	3,6
160	50	160,06	49,83	3,6
160	70	160,06	69,67	3,6
160	100	160,06	99,46	3,6
160	150	160,06	149,20	3,6

3 REMARKS

3.1 The reported uncertainties of measurement were calculated and expressed in accordance with the BIPM, IEC, ISO, IUPAP, and OIML document entitled "A Guide to the Expression of Uncertainty in Measurement" (International Organisation for Standardisation, Geneva, Switzerland).

3.2 The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by a coverage factor of k = 2, which for a normal distribution approximates a level of confidence of 95,45 %.

3.3 Certain of the NMISA certificates are consistent with the capabilities that are included in Appendix C of the MRA (Mutual Recognition Arrangement) drawn up by the CIPM. Under the MRA, all participating institutes recognise the validity of each other's calibration and measurement certificates for the quantities, ranges and measurement uncertainties specified in Appendix C. For details see https://www.bipm.org.

3.4 The calibrations were carried out at an ambient temperature of 23 °C ± 2 °C and a relative humidity between 35 %rh and 65 %rh.

3.5 The results reported relate only to the items calibrated.

3.6 NMISA is empowered by the Measurement Unit and Measurement Standards Act, Act No. 18 of 2006 and may issue a certificate stating the outcome of a calibration, measurement, or analysis. In any criminal proceedings a document which purports to be a certificate issued and signed in accordance with this act must upon its production be regarded as evidence of the facts contained therein.

3.7 Any mention of a third-party name on NMISA's certificates or any document, either through a request by the customer or through other instruction, shall not create an impression that NMISA is extending liability to that third party.

3.8 NMISA certificates are published on the NMISA portal and can be accessed by the registered customer only. Paper copies of this certificate can be issued on request at an additional fee as determined at the discretion of NMISA.

3.9 This certificate contains a QR code on the front page which reference the URL where authenticity of this certificate can be confirmed.

3.10 Certificates or reports may be freely published by the customer provided that such publication is verbatim and in full.

3.11 The measurement results recorded in this certificate were correct at the time of calibration. The subsequent accuracy will depend on factors such as care, handling and frequency of use. It is recommended that recalibration be undertaken at an interval that will ensure that the instrument remains within the desired limits.

---------- *end of certificate* ----------